BANK LOAN QUERY DOCUMENT

A. BANK LOAN REPORT | SUMMARY

1)Total Loan Applications

SELECT COUNT(id) as Total_Loan_Applications FROM bank_loan_data

Results | Messages

	Total_Loan_Applications
1	38576

2)MTD Loan application

SELECT COUNT(id) AS MTDTotal_Applications FROM bank_loan_data WHERE MONTH(issue_date) = 12

Results | Messages

	MTD_Total_Loan_Applications
1	4314

3) PMTD Loan Applications

SELECT COUNT(id) as PMTD_Total_Loan_Applications FROM bank_loan_data WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021

Results | Messages

	PMTD_Total_Loan_Applications
1	4035

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4)Total Funded Amount

SELECT SUM(loan_amount) as Total_Funded_Amount FROM bank_loan_data

Results | Messages

	Total_Funded_Amount
1	435757075

5)MTD Total Funded Amount

SELECT SUM(loan_amount) as MTDTotal_Funded_Amount FROM bank_loan_data WHERE MONTH(issue_date)= 12 AND YEAR(issue_date) = 2021

	MTDTotal_Funded_Amount
1	53981425

6)PMTD Total Funded Amount

SELECT SUM(loan_amount) as MTDTotal_Funded_Amount FROM bank_loan_data WHERE MONTH(issue_date)= 11 AND YEAR(issue_date) = 2021

	PMTDTotal_Funded_Amount
1	47754825

7)Total Amount Received

```
SELECT SUM(total_payment) as  Total_Amount_Received FROM bank_loan_data
```

	Total_Amount_Received
1	473070933

8)MTD Total Amount Received

```
SELECT SUM(total_payment) as  MTDTotal_Amount_Received FROM bank_loan_data WHERE
MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

Results | Messages

	MTDTotal_Amount_Received
1	58074380

9)PMTD Total Amount Received

```
SELECT SUM(total_payment) as  PMTDTotal_Amount_Received FROM bank_loan_data WHERE
MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTDTotal_Amount_Received
1	50132030

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9)Total Average Interest Rate Received

```
SELECT ROUND(AVG(int_rate),4)*100 as Average_Interest_Rate FROM bank_loan_data
```

Results | Messages

	Average_Interest_Rate
1	12.05

10)MTDTotal Average Interest Rate Received

```
SELECT ROUND(AVG(int_rate),4) *100 as MTDAverage_Interest_Rate FROM bank_loan_data WHERE
MONTH(issue_date)= 12 AND YEAR(issue_date) = 2021
```

Results | Messages

	Average_Interest_Rate
1	12.05

11)PMTDTotal Average Interest Rate Received

```
SELECT ROUND(AVG(int_rate),4) *100 as PMTDAverage_Interest_Rate FROM bank_loan_data WHERE
MONTH(issue_date)= 11 AND YEAR(issue_date) = 2021
```

	PMTDAverage_Interest_Rate
1	11.94

12)Average DTI

```
SELECT ROUND(AVG(dti),4) * 100  as Average_DTI from bank_loan_data
```

	Average_DTI
1	13.33

13)MTD Average DTI

```
SELECT ROUND(AVG(dti),4) * 100  as Average_DTI from bank_loan_data WHERE MONTH(issue_date) =
12 AND YEAR(issue_date) = 2021
```

Results | Messages

	MTDAverage_DTI
1	13.67

14)PMTD Average DTI

```
SELECT ROUND(AVG(dti),4) * 100  as Average_DTI from bank_loan_data WHERE MONTH(issue_date) =
11 AND YEAR(issue_date) = 2021
```

Results | Messages

	PMTDAverage_DTI
1	13.3

GOOD LOAN ISSUED

15)Good Loan Percentage

```
SELECT (COUNT(CASE  WHEN loan_status IN ('Fully paid','Current') THEN id  END) * 100) / COUNT(id)  AS
Good_loan_percentage

FROM bank_loan_data;
```

	Good_loan_percentage
1	86

16)Good Loan Applications

```
SELECT COUNT(id) as Good_Loan_Applications  FROM bank_loan_data WHERE loan_status IN ('Fully
Paid','Current')
```

Results | Messages

	Good_Loan_Applications
1	33243

17)Total Funded Amount In Good Loan Applications

```
SELECT SUM(loan_amount) AS Good_Loan_Funded_Amount  FROM bank_loan_data WHERE loan_status
IN ('Fully Paid','Current')
```

	Good_Loan_Funded_Amount
1	370224850

18)Good Loan Total Received Amount

```
SELECT SUM(total_payment) AS Good_Loan_Received_Amount  FROM bank_loan_data  WHERE
loan_status IN ('Fully Paid','Current')
```

	Good_Loan_Received_Amount
1	435786170

BAD LOAN ISSUED

19)Bad Loan Percentage

```
SELECT (COUNT(CASE  WHEN loan_status = 'Charged Off' THEN id  END)* 100.0) / COUNT(id) AS
Bad_loan_percentage FROM bank_loan_data;
```

	Bad_loan_percentage
1	13.824657818332

20)Bad Loan Applications

```
SELECT COUNT(id) as Bad_loan_applications FROM bank_loan_data  WHERE loan_status = 'Charged Off'
```

	Bad_loan_applications
1	5333

21)Bad Loan Funded Amount

```
SELECT SUM(loan_amount) as Bad_loan_Funded_Amount FROM bank_loan_data WHERE loan_status =
'Charged Off'
```

	Bad_loan_Funded_Amount
1	65532225

22)Bad Loan Funded Amount Received

```
SELECT SUM(total_payment) as Bad_loan_Amount_Received FROM bank_loan_data WHERE loan_status
= 'Charged Off'
```

	Bad_loan_Amount_Received
1	37284763

LOAN STATUS

```
SELECT loan_status,   COUNT(id) as Total_Loan_Applications ,  SUM(total_payment) AS
Total_Amount_Received,
        SUM(loan_Amount) AS Total_Funded_Amount,   AVG(int_rate * 100) As Interest_Rate,
        AVG(dti * 100 ) AS DTI
         FROM bank_loan_data
         GROUP BY loan_status
```

	loan_status	Total_Loan_Applications	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
1	Fully Paid	32145	411586256	351358350	11.6410707918092	13.1673507557434
2	Charged Off	5333	37284763	65532225	13.8785749318289	14.0047328005517
3	Current	1098	24199914	18866500	15.0993260800947	14.7243442736843

23) MTD Funded Amount, Amount Received

```
SELECT loan_status , SUM(loan_amount) as MTDFunded_Amount, SUM(total_payment) as
MTDAmount_Received FROM bank_loan_data WHERE MONTH(issue_date) = 12
group by loan_status
```

	loan_status	MTD_Funded_Amount	MTD_Amount_Received
1	Fully Paid	41302025	47815851
2	Charged Off	8732775	5324211
3	Current	3946625	4934318

24) PMTD Funded Amount, Amount Received

```
SELECT loan_status , SUM(loan_amount) as PMTDFunded_Amount, SUM(total_payment) as
MTDAmount_Received FROM bank_loan_data WHERE MONTH(issue_date) = 11
group by loan_status
```

Results | Messages

	loan_status	PMTDFunded_Amount	PMTDAmount_Received
1	Fully Paid	37375675	42420451
2	Charged Off	7511175	3994065
3	Current	2867975	3717514

Click to select the whole column

B. BANK LOAN REPORT | OVERVIEW

1)Monthly Trend

```
SELECT MONTH(issue_date) as Month_Number, DATENAME(MONTH,issue_date) as Month_Name,
COUNT(id) as Total_Loan_Applications,
SUM(loan_amount) as Total_Funded_Amount,
SUM(total_payment) as Total_Received_Amount
FROM bank_loan_data
GROUP BY MONTH(issue_date), DATENAME(MONTH,issue_date)
ORDER BY MONTH(issue_date)
```

Results | Messages

	Month_Number	Month_Name	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

2)Regional Analysis By Trend

```
SELECT address_state, COUNT(id) as Total_Loan_Applications, SUM(loan_amount) as
Total_Funded_Amount, SUM(total_payment) as Total_Received_Amount
FROM bank_loan_data
GROUP BY address_state
ORDER BY SUM(loan_amount) DESC
```

Results | Messages

	address_state	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	CA	6894	78484125	83901234
2	NY	3701	42077050	46108181
3	TX	2664	31236650	34392715
4	FL	2773	30046125	31601905
5	NJ	1822	21657475	23425159
6	IL	1486	17124225	18875941
7	VA	1375	15982650	17711443
8	PA	1482	15826525	17462908
9	GA	1355	15480325	16728040
10	MA	1310	15051000	16676279
11	OH	1188	12991375	14330148
12	MD	1027	11911400	12985170
13	AZ	833	9206000	10041986
14	CO	770	8976000	9845810
15	WA	805	8855525	9531739
16	NC	759	8787575	9534813
17	CT	730	8435575	9357612
18	MI	685	7829900	8543660

3)Loan term analysis

```
SELECT term, COUNT(id) as Total_Loan_Applications, SUM(loan_amount) as Total_Funded_Amount,
SUM(total_payment) as Total_Received_Amount
FROM bank_loan_data
GROUP BY term
ORDER BY term
```

	term	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

4)Employee Length Analysis

```
SELECT emp_length, COUNT(id) as Total_Loan_Applications, SUM(loan_amount) as Total_Funded_Amount,
SUM(total_payment) as Total_Received_Amount
FROM bank_loan_data
GROUP BY emp_length
ORDER BY emp_length
```

	emp_length	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	< 1 year	4575	44210625	47545011
2	1 year	3229	32883125	35498348
3	10+ years	8870	116115950	125871616
4	2 years	4382	44967975	49206961
5	3 years	4088	43937850	47551832
6	4 years	3428	37600375	40964850
7	5 years	3273	36973625	40397571
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

5)Loan Purpose

```
SELECT purpose, COUNT(id) as Total_Loan_Applications, SUM(loan_amount) as Total_Funded_Amount,
SUM(total_payment) as Total_Received_Amount FROM bank_loan_data
GROUP BY purpose
ORDER BY purpose
```

	purpose	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	car	1497	10223575	11324914
2	credit card	4998	58885175	65214084
3	Debt consolidation	18214	232459675	253801871
4	educational	315	2161650	2248380
5	home improvement	2876	33350775	36380930
6	house	366	4824925	5185538
7	major purchase	2110	17251600	18676927
8	medical	667	5533225	5851372
9	moving	559	3748125	3999899
10	other	3824	31155750	33289676
11	renewable_energy	94	845750	898931
12	small business	1776	24123100	23814817
13	vacation	352	1967950	2116738
14	wedding	928	9225800	10266856

6)Home Ownership

```
SELECT home_ownership, COUNT(id) as Total_Loan_Applications, SUM(loan_amount) as
Total_Funded_Amount, SUM(total_payment) as Total_Received_Amount
FROM bank_loan_data
GROUP BY home_ownership
ORDER BY count(id) DESC
```

	home_ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	RENT	18439	185768475	201823056
2	MORTGAGE	17198	219329150	238474438
3	OWN	2838	29597675	31729129
4	OTHER	98	1044975	1025257
5	NONE	3	16800	19053